

$^{71}\text{Ga}({}^3\text{He,t}),({}^3\text{He,t}\gamma)$ 2015Fr02,1998Ej03,1988Ch25

Type	Author	History	Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh and Jun Chen		NDS 188,1 (2023)	17-Jan-2023

Target $J^\pi=3/2^-$.

The $^{71}\text{Ga}({}^3\text{He,t})$ studies are relevant to determination of neutrino capture rates in ^{71}Ga , a detector of solar neutrino flux.

2015Fr02 (also **2011Fr15**): (${}^3\text{He,t}$), $E=420$ MeV (450 MeV in **2011Fr15**). Measured triton spectra, and angular distributions using Grand Raiden spectrometer with FWHM=45 keV at RCNP, Osaka cyclotron facility. DWBA analysis of $\sigma(\theta)$ data. Deduced $B(>)$ and solar-neutrino capture rates.

1998Ej03 (also **1996Fu06,1994Fu11**): (${}^3\text{He,t}$), (${}^3\text{He,t}\gamma$), $E=450$ MeV from RCNP ring cyclotron. Measured triton energies at $0-2^\circ$ in the forward direction in singles and coin with γ rays. Tritons were analyzed by GRAND RAIDEN magnetic spectrometer with FWHM=300-400 keV. The γ rays were detected using six NaI(Tl) detectors with FWHM=100-300 keV. The target was enriched and 2 mg/cm² thick. The CASCADE and GEANT codes were used to evaluate and simulate γ -spectrum. Deduced Gamow-Teller strengths and solar neutrino capture rates.

1988Ch25: (${}^3\text{He,t}\gamma$), (${}^3\text{He,t}$): $E=29.9$ MeV. Measured $E(t)$ in singles (at 0°) and coin with γ rays using Q3D magnetic spectrometer for tritons and NaI(Tl) detector for γ rays. 99.8% enriched target. Detailed study of the isobaric analog state at 8.9 MeV.

1984Ko10 (from the same group as **1988Ch25**): (${}^3\text{He,t}$) $E=20.42$ MeV. Measured triton spectra using Q3D spectrometer, deduced Q value and excited states in ^{71}Ge .

Not all the peaks shown in triton spectra of **1998Ej03** and **1984Ko10** are identified.

 ^{71}Ge Levels

$B(>)=2.88$ 9 (**1998Ej03**) for 1580-7420 energy region.

$B(\text{GT})$: From **2015Fr02**.

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$\beta(>)$	Comments
0	$1/2^-$	0.0852 11	$d\sigma/d\Omega(q=0)=0.786$ mb/sr 9 (2015Fr02). Solar neutrino capture rate=109.8 SNU 13 (2015Fr02). $B(>)=0.0891$ 13 (1998Ej03).
175 1	$5/2^-$	0.0034 26	$d\sigma/d\Omega(q=0)=0.071$ mb/sr 4 (2015Fr02). Solar neutrino capture rate=1.2 SNU 7 (2015Fr02). $B(>)=0.0049$ 18 (1998Ej03).
500 1	$3/2^-$	0.0176 14	$d\sigma/d\Omega(q=0)=0.171$ mb/sr 4 (2015Fr02). Solar neutrino capture rate=2.7 SNU 2 (2015Fr02). $B(>)=0.0208$ 21 (1998Ej03).
708 1	$3/2^-$	0.0011 5	$d\sigma/d\Omega(q=0)=0.018$ mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.03 SNU 1 (2015Fr02).
808 1	$1/2^-$	0.0229 10	$E(\text{level})$: 830 in 1998Ej03 . $d\sigma/d\Omega(q=0)=0.210$ mb/sr 4 (2015Fr02). Solar neutrino capture rate=0.61 SNU 7 (2015Fr02). $B(>)=0.0237$ 23 (1998Ej03).
1096 1	$3/2^-$	0.0183 17	$d\sigma/d\Omega(q=0)=0.184$ mb/sr 4 (2015Fr02). Solar neutrino capture rate=0.33 SNU 3 (2015Fr02). $B(>)=0.0233$ 24 (1998Ej03).
1299 1	$3/2^-$	0.0133 8	$E(\text{level})$: 1096 (1984Ko10), 1160 (1998Ej03). $d\sigma/d\Omega(q=0)=0.126$ mb/sr 2 (2015Fr02). Solar neutrino capture rate=0.17 SNU 1 (2015Fr02). $B(>)=0.0201$ 23 (1998Ej03).
1378 1	$5/2^-$	0.0033 4	$E(\text{level})$: 1299 (1984Ko10), 1360 in 1998Ej03 could be a composite of 1299 and 1378 keV peaks. $d\sigma/d\Omega(q=0)=0.035$ mb/sr 3 (2015Fr02). Solar neutrino capture rate=0.041 SNU 5 (2015Fr02).
1598 1	$3/2^-$	0.0011 5	$d\sigma/d\Omega(q=0)=0.018$ mb/sr 2 (2015Fr02). Solar neutrino capture rate=0.013 SNU 6 (2015Fr02).
1744 1	$3/2^-$	0.0068 2	$d\sigma/d\Omega(q=0)=0.061$ mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.077 SNU 3 (2015Fr02).

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$^{71}\text{Ga}({}^3\text{He,t}),({}^3\text{He,t}\gamma)$ **2015Fr02,1998Ej03,1988Ch25** (continued) ^{71}Ge Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	β(>)	Comments
1964 1	3/2 ⁻		0.0012 6	dσ/dΩ(q=0)=0.020 mb/sr 2 (2015Fr02). Solar neutrino capture rate=0.013 SNU 6 (2015Fr02).
2041 1	(3/2 ⁻ ,5/2 ⁻)		0.0181 2	dσ/dΩ(q=0)=0.187 mb/sr 3 (2015Fr02). Solar neutrino capture rate=0.187 SNU 6 (2015Fr02).
2145 1	(3/2 ⁻ ,5/2 ⁻)		0.0031 6	dσ/dΩ(q=0)=0.036 mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.031 SNU 6 (2015Fr02).
2276 1	(3/2 ⁻ ,5/2 ⁻)		0.0036 9	dσ/dΩ(q=0)=0.046 mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.035 SNU 9 (2015Fr02).
2352 1	(3/2 ⁻ ,5/2 ⁻)		0.0130 28	dσ/dΩ(q=0)=0.157 mb/sr 2 (2015Fr02). Solar neutrino capture rate=0.12 SNU 3 (2015Fr02).
2435 1	(3/2 ⁻ ,5/2 ⁻)		0.0123 17	dσ/dΩ(q=0)=0.133 mb/sr 2 (2015Fr02). Solar neutrino capture rate=0.11 SNU 2 (2015Fr02).
2642 1	(5/2 ⁻)		0.0054 10	dσ/dΩ(q=0)=0.062 mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.046 SNU 9 (2015Fr02).
2778 1	(5/2 ⁻)		0.0058 12	dσ/dΩ(q=0)=0.070 mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.048 SNU 10 (2015Fr02).
2806 1	(5/2 ⁻)		0.0172 12	dσ/dΩ(q=0)=0.165 mb/sr 3 (2015Fr02). Solar neutrino capture rate=0.140 SNU 11 (2015Fr02).
2888 1	(5/2 ⁻)		0.0019 7	dσ/dΩ(q=0)=0.028 mb/sr 4 (2015Fr02). Solar neutrino capture rate=0.015 SNU 6 (2015Fr02).
2924 1	(5/2 ⁻)		0.0033 14	dσ/dΩ(q=0)=0.052 mb/sr 1 (2015Fr02). Solar neutrino capture rate=0.026 SNU 11 (2015Fr02).
325×10 ¹ 25			0.110 10	E(level): 3.0-3.5 MeV energy range. Solar neutrino capture rate=0.77 SNU 7 (2015Fr02).
375×10 ¹ 25			0.165 16	E(level): 3.5-4.0 MeV energy range. Solar neutrino capture rate=0.96 SNU 10 (2015Fr02).
425×10 ¹ 25			0.191 17	E(level): 4.0-4.5 MeV energy range. Solar neutrino capture rate=0.92 SNU 9 (2015Fr02).
475×10 ¹ 25			0.209 16	E(level): 4.5-5.0 MeV energy range. Solar neutrino capture rate=0.82 SNU 7 (2015Fr02).
525×10 ¹ 25			0.191 18	E(level): 5.0-5.5 MeV energy range. Solar neutrino capture rate=0.61 SNU 6 (2015Fr02).
575×10 ¹ 25			0.265 19	E(level): 5.5-6.0 MeV energy range. Solar neutrino capture rate=0.67 SNU 5 (2015Fr02).
625×10 ¹ 25			0.338 18	E(level): 6.0-6.5 MeV energy range. Solar neutrino capture rate=0.66 SNU 4 (2015Fr02).
675×10 ¹ 25			0.315 17	E(level): 6.5-7.0 MeV energy range. Solar neutrino capture rate=0.47 SNU 3 (2015Fr02).
721×10 ¹ 21			0.289 16	E(level): 7.0-7.42 MeV energy range. Solar neutrino capture rate=0.33 SNU 2 (2015Fr02).
794×10 ¹ 52			0.645 39	E(level): 7.42-8.46 MeV energy range. B(>)=0.74 10 (1998Ej03). %n≈100 (1988Ch25) B(Fermi)=0.0900 22 (2011Fr15). 53% 9 decay proceeds through neutron decay to 1.22 MeV state in ^{70}Ge with subsequent observation of 1040γ in ^{70}Ge . The γ decay of IAS to ^{71}Ga through single high-energy γ ray or a cascade of two γ rays is determined to be less than 11% (1988Ch25). E(level): from 2011Fr15 and 2015Fr02, with uncertainty estimated as 5 keV by evaluators based on FWHM=45 keV. Other: 8932 13 (1988Ch25). J ^π : IAS of 3/2 ⁻ g.s. in ^{71}Ga . Γ from 2011Fr15. dσ/dΩ(q=0)=9.04 mb/sr 12 (2011Fr15).
8913 5	3/2 ⁻	≈50 keV		

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$^{71}\text{Ga}({}^3\text{He,t}),({}^3\text{He,t}\gamma)$ [2015Fr02,1998Ej03,1988Ch25](#) (continued)

^{71}Ge Levels (continued)

<u>E(level)[†]</u>	<u>Comments</u>
11.75×10 ³	Gamow-Teller giant resonance (GTGR), wide structure from 10-16 MeV, peaking at ≈11.75 MeV (2015Fr02,2011Fr15).
18.0×10 ³	Interpreted as T _{>} component of the GTGR (2015Fr02,2011Fr15).

[†] From [2015Fr02](#), unless otherwise noted.

[‡] States populated by Gamow-Teller excitations are expected to be 1/2⁻, 3/2⁻ or 5/2⁻. J^π assignments are as given in Table I of [2015Fr02](#), which are based on literature assignments for low-lying levels, and from $\sigma(\theta)$ data with comparison to DWBA calculations in the higher energy region.